Section 3.3 Derivatives of Trig Functions

Learn these 3 derivatives first.

$$(\sin x)' = \cos x$$

$$(\tan x)' = \sec^2 x$$

$$(\sec x)' = \sec x \cdot \tan x$$

Then it will be easy to remember the next 3 derivatives.

$$(\cos x)' = -\sin x$$

$$(\cot x)' = -\csc^2 x$$

$$(\csc x)' = -\csc x \cdot \cot x$$

You will need to remember all 6 of these, so do whatever it takes to get this done as soon as you can.

They can also be organized like this:

$(\sin x)' = \cos x$ $\qquad$ $(\cos x)' = -\sin x$

$(\tan x)' = \sec^2 x$ $\qquad$ $(\cot x)' = -\csc^2 x$

$(\sec x)' = \sec x \cdot \tan x$ $\qquad$ $(\csc x)' = -\csc x \cdot \cot x$

Now we can use these together with the other rules we've learned.

Ex ① $f(x) = e^x \cdot \sin x$ use Product Rule

$f'(x) = e^x \cdot \sin x + e^x \cdot \cos x$

or $f'(x) = e^x(\sin x + \cos x)$ ✓

② $y = \cos^2 x$ write this as a product

$y = \cos x \cdot \cos x$ now use Product Rule

$$y' = -\sin x \cdot \cos x + \cos x \cdot (-\sin x)$$

or $$y' = -2\sin x \cdot \cos x \quad \checkmark$$

Note for $y = \cos^2 x$ we cannot just use the Power Rule!

The reason for this is that $\cos^2 x$ means $(\cos x)^2$ which is the composition of two functions. More about this in the next section.

Ex ③ $h(x) = \dfrac{\sec x}{x}$ use the Quotient Rule

$$h'(x) = \frac{\sec x \cdot \tan x \cdot x - 1 \cdot \sec x}{x^2}$$

or $h'(x) = \dfrac{\sec x \cdot (x \cdot \tan x - 1)}{x^2}$ ✓

Ex ④ Find the equation of the tangent line for $f(x) = x \cdot \cos x$ when $x = 0$.

first find the y-value

$$f(0) = 0 \cdot \cos(0) = 0$$

So, the point is $(0,0)$.

now use the Product Rule

$$f'(x) = 1 \cdot \cos x + x \cdot (-\sin x)$$

plug in $x=0$ to find the slope

$$f'(0) = \cos(0) + 0(-\sin(0))$$

$$\text{slope} = 1$$

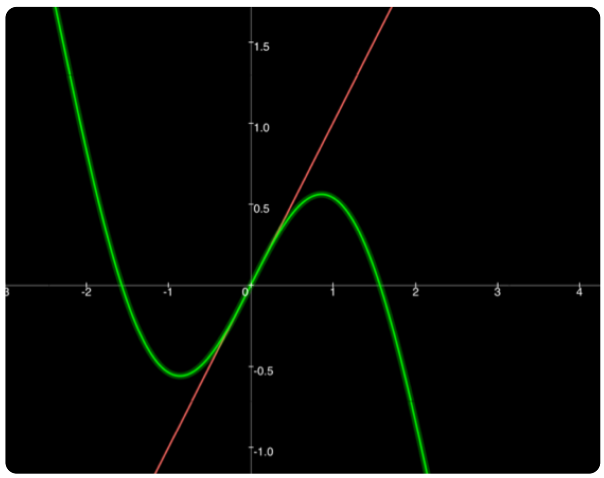
use the point-slope formula

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x - 0)$$

$\boxed{y = x}$ is the tangent line at $(0,0)$

Now graph the original function and $y=x$ to see if the tangent line looks right.



That looks right!

Ex ⑤ Can $y' = 0$ for $y = \tan x$?

$y' = \sec^2 x$ set it equal to 0

$0 = \sec^2 x$ use identity $\sec x = \frac{1}{\cos x}$

$0 = \frac{1}{\cos^2 x}$ multiply both sides by $\cos^2 x$

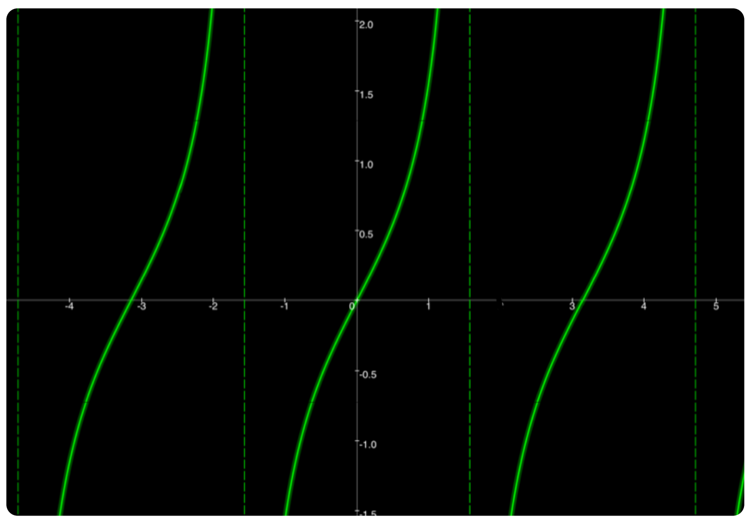
$0 = 1$ so, there is no solution.

But notice that $\frac{1}{\cos^2 x}$ is undefined

when $\cos^2 x = 0$. NOTE $\cos\left(n \cdot \frac{\pi}{2}\right) = 0$

This means that something important is happening to $y = \tan x$ at $n \cdot \frac{\pi}{2}$.

$y = \tan x$ has vertical asymptotes at $n \cdot \frac{\pi}{2}$



Ex ⑥ Prove that $(\sec x)' = \sec x \cdot \tan x$

use $\sec x = \frac{1}{\cos x}$ and the Quotient Rule

$$(\sec x)' = \left(\frac{1}{\cos x}\right)'$$

$$= \frac{0 \cdot \cos x - 1 \cdot (-\sin x)}{\cos^2 x}$$

$$= \frac{1 \cdot \sin x}{\cos x \cdot \cos x}$$

$$= \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x}$$

$$= \sec x \cdot \tan x$$